



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)



Project Synopsis

Batch No: B3

Date: 26th July 26, 2025

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Project title:

Introduction:

In today's educational environments, efficient use of infrastructure like classrooms plays a critical role in ensuring smooth academic operations. Often, students and faculty face difficulty in locating available classrooms during unplanned sessions or events. This leads to confusion, wasted time, and underutilized resources. With the increasing emphasis on digital transformation, there is a growing need for systems that can monitor classroom usage and provide real-time availability information. This project aims to address such challenges by developing a Classroom Availability Tracker using a combination of frontend, backend, and database technologies, with hardware support for real-time updates. The idea is to bring a smarter, more transparent way to manage classroom availability in institutions.

Objectives:

The primary objective of this project is to develop a robust and user-friendly system that allows authorized faculty members to update the occupancy status of a classroom securely.

It aims to:

- Authenticate faculty identity using a fingerprint sensor.
- Provide a single-button interface for toggling classroom occupancy status.
- Display real-time status messages on a compact OLED screen.
- Store updates in a remote database for centralized monitoring.
- Ensure that the room status is automatically synced with the MERN web application.
- Ensure that only authenticated personnel can alter room availability.
- Enable institutions to scale this solution across multiple classrooms with minimal cost and effort.

Software Requirements:

- **Arduino IDE:** For writing and uploading code to the ESP32 microcontroller.
- **MERN Stack:** Manages web interface, data APIs, and storage of classroom status.
- **MongoDB:** Cloud-based NoSQL database storing room states, user fingerprints and timestamps.
- **JWT Authentication:** Secures backend API access and restricts room status updates to authenticated faculty.
- **Postman:** Used for testing and debugging HTTP requests between hardware and backend.
- **Wi-Fi Network:** Required for the ESP32 to connect and send updates to the backend server.

Hardware Requirements

ESP32 Microcontroller: Acts as the central controller for the fingerprint sensor, buttons, and OLED display. It also handles Wi-Fi communication to sync with the backend.

R307 Fingerprint Sensor: Captures and authenticates fingerprints of authorized faculty members.

OLED Display (0.96" I2C): Displays the room status and messages like "Access Granted."

Push Buttons (2): Used by authenticated faculty to manually select the current room status

Power Source: 5V USB power bank or adapter to power the ESP32 and peripheral modules.

Jumper Wires & Breadboard: For making prototype connections between ESP32 and peripherals.

Abstract:

This project introduces a smart classroom occupancy tracker that integrates fingerprint authentication and real-time status reporting. Faculty members use their fingerprint to access a simple button interface through which they can set the classroom status. An OLED display provides immediate feedback to confirm the action, and the change is communicated over Wi-Fi to a MERN-based web app, where students and staff can check current room availability. This fusion of embedded systems and full-stack development aims to create a transparent, user-friendly, and secure solution to classroom allocation and monitoring.

Literature Survey (Minimum 5 Paper):

Several educational institutions have tried to address classroom management challenges through scheduling software and manual systems. Tools like Google Sheets or Excel-based trackers are still commonly used in many colleges. While some universities have adopted ERP systems with built-in room tracking, these systems are often costly, complex, and not customizable for small institutions. Research papers suggest that the integration of IoT with web-based systems has potential in solving real-time occupancy problems, especially in smart campuses. Systems using PIR sensors, RFID attendance, and camera-based monitoring have shown good accuracy but require substantial infrastructure. Compared to these, our approach provides a practical balance between automation and usability by leveraging web technologies and affordable hardware.

Existing System & its Drawback:

- Availability is shown manually on paper or whiteboards outside classrooms.
- Faculty must physically check rooms or contact others to confirm availability.
- Manual updates cause delays, errors, and lack real-time accuracy.
- Static timetables don't reflect last-minute changes or cancellations.
- No authentication—room status can be changed by anyone.
- No centralized online platform for students or staff to check room status.

Proposed System & Advantages:

- Uses ESP32 with fingerprint sensor for secure faculty authentication.
- Allows faculty to toggle room status via button press (Vacant/Occupied).
- OLED screen displays real-time status feedback in the classroom.
- Updates are sent instantly to a central database via Wi-Fi.
- MERN-based web app lets users check availability from any device.
- Improves accuracy, security, accessibility, and reduces manual effort.

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